

A Hybrid Recommendation System Combining Collaborative Filtering, Sentiment Analysis, and Content-Based Features for Review-Based Rating Prediction

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Abstract—Recommender systems play an important role in modern e-commerce, although conventional collaborative filtering (CF), particularly matrix factorization, can be affected by sparse user-item interactions and imbalanced rating distributions. A hybrid framework is developed by extending a Singular Value Decomposition (SVD) baseline with review-text features, including a VADER sentiment compound score and dimensionality-reduced TF-IDF representations, which are integrated using a Random Forest meta-learner. Experiments are conducted using a stratified sample of 100,001 Amazon Fine Food Reviews with a shared 70/15/15 train/validation/test split. Hyperparameters are selected through validation-set grid search. The same-review hybrid produces an RMSE of 0.9604 compared with 1.0068 for the tuned SVD baseline, corresponding to a 4.61% reduction in RMSE and a 19.71% reduction in MAE. A leakage-free, cold-start-safe configuration uses leave-one-out aggregates derived from other reviews of the same product. With this configuration, the RMSE improvement decreases to 0.69%, while the MAE improvement remains at 15.40%. These findings show that aggregated review content can provide predictive value beyond collaborative filtering, although the RMSE benefit is considerably reduced when same-review information is unavailable. Feature-importance analysis indicates that the SVD prediction accounts for more than 95% of the Random Forest decision weight in both configurations. The hybrid achieves better performance for 1, 2, and 5-star ratings, whereas SVD performs better for 3 and 4-star ratings. Precision@K results remain broadly similar, largely due to the dataset’s 62% positive-rating skew, which restricts the effectiveness of ranking-based evaluation.

Index Terms—Recommender systems, collaborative filtering, hybrid recommendation, sentiment analysis, matrix factorization, TF-IDF, Random Forest, Precision@K, Amazon reviews.

I. INTRODUCTION

The rapid growth of e-commerce has resulted in review datasets containing hundreds of millions of records, creating both opportunities and challenges for recommender systems. While rating information remains relatively sparse across the extensive space of users and items, the corresponding review text provides additional information that cannot be captured by rating matrices alone [1], [2]. Collaborative filtering (CF), particularly matrix factorization techniques such as Singular Value Decomposition (SVD), remains a strong and widely

used baseline for rating prediction [3]–[5]. However, collaborative filtering methods can face limitations in sparse interaction and cold-start scenarios. When adequate interaction history is not available, predictions may depend substantially on global, user-level, or item-level average estimates [6], [7].

As a result, considerable research has focused on incorporating review text into collaborative filtering. Textual information has been employed to identify latent rating dimensions [8], capture changes in reviewer expertise over time [9], and directly integrate rating and textual information within prediction models [10], [11]. More recent methods have further developed this integration using neural architectures [12], [13], while wider research has explored the development of deep hybrid recommender systems [14], [15].

This study examines an empirical hybrid recommendation pipeline that combines collaborative filtering with information derived from review text. The methodology initially cleans and filters the Amazon Fine Food Reviews dataset to address sparse user-item interactions. Exploratory data analysis is then conducted to investigate rating distributions, review lengths, and user and product activity. An SVD collaborative filtering model and a hybrid Random Forest model are subsequently trained using the same canonical train/validation/test split. Using a shared split ensures that both approaches are assessed using identical held-out observations and prevents sampling differences from influencing the comparison. Hyperparameters for both models are selected using the validation partition, followed by final evaluation on the untouched test partition.

The review-text component generates two categories of features. First, a VADER sentiment compound score is obtained from the review text [16]. Second, the review content is represented using TF-IDF and then reduced in dimensionality. These textual features are combined with the SVD prediction and provided to a Random Forest meta-learner. Model performance is assessed using RMSE and MAE for rating prediction, together with Precision@K under two configurations developed to account for the severe per-user sparsity within the dataset [6], [17].

Alongside the aggregate performance measures, analysis is performed for individual rating classes to determine the categories in which the hybrid model achieves improvements or generates larger errors than the SVD baseline. This provides a more detailed assessment of model behaviour than aggregate metrics alone. The evaluation also considers the use of review information from the same review for which the rating is being predicted. Since this information would not ordinarily be available when making a recommendation or predicting a rating, a separate leakage-free, cold-start-safe configuration is introduced using leave-one-out aggregates from other reviews associated with the same product.

Comparing the two feature-construction strategies provides a direct assessment of the contribution of review content under more realistic conditions. The same-review configuration achieves a 4.61% reduction in RMSE and a 19.71% reduction in MAE compared with the tuned SVD-only baseline, while the leakage-free configuration achieves a 0.69% reduction in RMSE and a 15.40% reduction in MAE.

The resulting analysis provides a more comprehensive interpretation of hybrid recommendation performance by considering model tuning, data leakage, rating-class behaviour, feature importance, and ranking-based evaluation rather than depending solely on a single overall error measure.

II. RELATED WORK

A. Collaborative Filtering and Matrix Factorization

Matrix factorization methods, particularly Singular Value Decomposition (SVD) and its regularized variants, have been extensively applied in collaborative filtering since the Netflix Prize period [3], [4]. These techniques represent users and items using latent factors that capture underlying patterns in rating interactions. Probabilistic formulations have further expanded the use of matrix factorization approaches [5], while dimensionality-reduction methods have introduced additional ways of modelling recommendation data [18]. Early research on recommender-system evaluation has also established several performance measures that remain widely used for evaluating collaborative filtering systems [6], [7].

Despite their effectiveness, matrix factorization methods rely heavily on the availability of user-item interaction information. Consequently, sparse interaction histories can negatively affect prediction quality, particularly for users or items with only a limited number of observed ratings. These limitations motivate the integration of additional information sources, such as review text, with rating-based collaborative filtering.

B. Review-Text-Aware Recommendation

A separate line of research has focused on incorporating textual review information directly into rating prediction. McAuley and Leskovec showed that latent topics derived from review text can represent hidden dimensions associated with rating behaviour [8]. Related research further demonstrated that review text can be used to capture changes in reviewer expertise over time [9]. These findings indicate that review

text can provide information about user preferences and item characteristics that may not be explicitly captured by rating matrices.

Other methods have integrated textual and rating information using graph-based and topic-based approaches. For example, TriRank incorporates information derived from reviews to support explainable recommendation [10], while rating-boosted topic models combine ratings with textual topics to represent users and items [11]. Together, these approaches demonstrate the potential value of incorporating review content alongside conventional rating-based recommendation methods.

C. Hybrid and Deep Learning Approaches

The development of hybrid recommendation approaches has also progressed through the use of neural network architectures. Neural Collaborative Filtering replaces the traditional inner-product interaction used in classical matrix factorization with a learned neural function [12]. TransNets further develops review-aware recommendation by learning transformations between representations derived from user and item reviews [13].

More broadly, surveys of recommender systems based on deep learning have investigated the application of neural architectures for integrating different information sources and modelling complex interactions between users and items [14]. Recent studies have also explored generative-model-based methods in recommender systems, demonstrating the continued development of increasingly advanced hybrid recommendation architectures [15].

Although neural and generative approaches offer more sophisticated mechanisms for integrating different recommendation signals, simpler hybrid architectures remain valuable for empirical evaluation. In particular, combining a conventional collaborative filtering prediction with explicitly derived textual features provides an interpretable framework for investigating the additional predictive value provided by review information.

D. Sentiment Analysis for Review Mining

Sentiment analysis offers an additional approach for extracting useful information from review text. VADER is a lexicon- and rule-based sentiment analysis method developed for short and informal text, including social media content and product reviews [16]. It is particularly suitable for review analysis because its scoring approach preserves sentiment-related cues conveyed through capitalization, punctuation, and exclamation marks, which may otherwise be removed during standard text-preprocessing procedures [16].

Accordingly, the present methodology applies VADER to the original review text when generating the sentiment feature. Broader natural language processing research provides further foundations for the representation and analysis of language data [19], while neural-based approaches have expanded the range of techniques available for processing textual information [20].

E. Evaluation Methodology

Evaluating recommender systems requires consideration of more than a single overall error metric. Different evaluation measures capture different aspects of recommendation quality, while examining specific user or rating segments can reveal patterns that may not be apparent from aggregate results alone [6], [17]. Dataset characteristics, such as sparsity and rating imbalance, can also affect how evaluation outcomes should be interpreted.

Recent research on recommender systems continues to highlight the importance of suitable evaluation procedures when comparing different recommendation approaches [21]–[23]. In sparse review datasets, ranking-based metrics can be particularly challenging to interpret because many users may have only a limited number of test interactions. Evaluation can therefore be strengthened by combining rating-prediction metrics with ranking-based measures while also considering the characteristics of the underlying dataset.

The methodology adopted in this study reflects this broader evaluation approach by using RMSE and MAE to assess rating prediction and Precision@K to evaluate ranking performance. In addition, per-rating-class analysis is conducted to determine whether the hybrid approach produces consistent improvements across different rating levels.

III. METHODOLOGY

Figure 1 illustrates the complete pipeline followed in the study. The raw review data are first cleaned and separated into the required data partitions. The processed data then proceed through two distinct branches: a collaborative-filtering branch that uses only the rating matrix and a content-based branch that extracts information from the review text. The outputs of these two branches are then combined to form the input features for the Random Forest meta-learner.

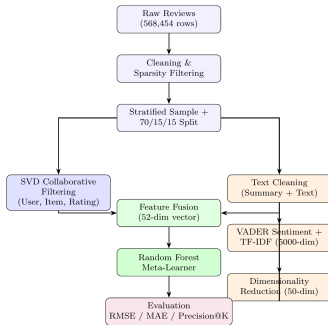


Fig. 1. End-to-end pipeline from raw Amazon Fine Food Reviews to the final hybrid rating prediction. Following cleaning, sampling, and the canonical train/validation/test split, the data follow two independent branches: a collaborative-filtering branch using only the rating matrix, and a content branch processing review text into VADER sentiment and dimensionality-reduced TF-IDF features. Both branches are fused into a 52-dimensional feature vector supplied to the Random Forest meta-learner.

A. Dataset and Preprocessing

The experiments use the Amazon Fine Food Reviews dataset, which contains 568,454 reviews across 10 original

fields [8], [9]. During preprocessing, ProductId, UserId, Score, Summary, and Text are retained. Rows with missing UserId, ProductId, Score, or Text values are removed, after which 892 exact duplicate rows are eliminated, leaving 567,562 reviews. To address sparse interactions, iterative filtering is applied with a minimum requirement of 3 reviews per user and 3 reviews per product. This filtering is repeated until the dataset stabilises, requiring 5 iterations and resulting in 275,930 reviews from 40,760 users and 13,753 products.

B. Exploratory Data Analysis

The cleaned dataset exhibits a strong positive-rating imbalance, with 62.18% of reviews receiving 5 stars and 16.16% receiving 4 stars. The remaining 21.66% consist of 1, 2, and 3-star ratings. Reviews contain an average of 87.6 words, with a median of 61 words and a maximum of 3,432 words. User activity is also highly skewed, with a median of 4 reviews, a mean of 6.77 reviews, and a maximum of 331 reviews per user. A similar pattern is observed for product activity, with a median of 6 reviews, a mean of 20.06 reviews, and a maximum of 631 reviews per product. These distributions are summarised in Figure 2.

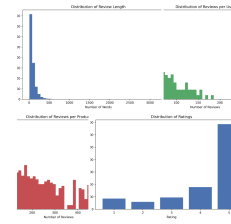


Fig. 2. Exploratory data analysis on the cleaned dataset (275,930 reviews). All four distributions are heavily right-skewed, with rating (a) skewed toward 5-star and both activity distributions (c, d) showing the long-tail pattern typical of review platforms.

C. Sampling and Canonical Train/Test Split

A stratified sample of 100,001 reviews is drawn from the cleaned dataset while maintaining the original rating distribution. The sample is divided once into 70,000 training rows, 15,001 validation rows, and 15,001 test rows, resulting in a 70/15/15 split stratified according to rating. This same canonical split is applied to all models to ensure that identical held-out test observations are used for comparison. The validation partition is used solely for hyperparameter selection and is excluded from the calculation of the reported test metrics.

D. Sentiment Feature Extraction

VADER sentiment analysis [16] is performed on the concatenated raw Summary and Text fields for each review. The raw text is preserved because VADER considers capitalization, punctuation, and exclamation marks as indicators of sentiment intensity, which may be removed during standard text-preprocessing. As a validation check, the mean compound sentiment increases consistently with the star rating, ranging from 0.017 for 1-star reviews to 0.861 for 5-star reviews.

E. Content Representation

The cleaned and combined review text is represented using TF-IDF with a 5,000-term vocabulary. The vocabulary and IDF weights are fitted only using training rows and subsequently applied to the validation and test rows to prevent held-out information from influencing the feature space. The resulting representation is reduced to 50 components using TruncatedSVD, fitted only on the training data, with 16.98% of the original variance retained.

F. Collaborative Filtering Baseline

The cleaned and combined review text is transformed into TF-IDF representations using a 5,000-term vocabulary. Both the vocabulary and IDF weights are fitted exclusively on the training rows and then applied to the validation and test rows to prevent information from the held-out data from affecting the feature space. The resulting representation is further reduced to 50 components using TruncatedSVD, which is also fitted only on the training data, retaining 16.98% of the original variance.

G. Hybrid Feature Fusion and Meta-Learner

The SVD prediction, VADER compound score, and 50-dimensional reduced TF-IDF representation are combined to form a 52-dimensional feature vector for each review. A Random Forest Regressor is subsequently trained using the same canonical training split. Its hyperparameters are determined through a validation-set grid search, after which the final model is evaluated using the same test split as the SVD baseline. This feature-fusion approach is consistent with the general hybrid recommendation strategy adopted in previous review-aware methods [10], [11].

H. Hyperparameter Tuning

Hyperparameters for both models are determined through a small grid search using the validation split, with the test data excluded from the tuning process. Four SVD configurations are assessed by varying the number of latent factors, training epochs, learning rate, and regularisation strength. The selected configuration uses 50 factors, 30 epochs, a learning rate of 0.01, and regularisation of 0.05. Four Random Forest configurations are also evaluated using different numbers of trees and maximum tree depths, with the selected configuration consisting of 200 trees and unlimited depth. Both selected models are then refitted using only the training split.

This tuning leads to a substantial improvement in the SVD baseline. The earlier untuned comparison reported an 11.03% RMSE improvement for the hybrid model, while tuning the SVD baseline reduces its test RMSE from 1.0903 to 1.0068. Consequently, the improvement achieved by the hybrid model becomes smaller, but the resulting comparison is more balanced and methodologically fair.

I. Evaluation Metrics

Rating-prediction performance is evaluated using RMSE and MAE. Recommendation performance is further assessed using Precision@K, with test items receiving a true rating of at least 4 treated as relevant [6], [7]. Precision@3 is computed across users with at least one relevant test item, whereas Precision@10 is evaluated only for users with at least 10 test items. Coverage is also reported alongside the stricter Precision@10 evaluation.

J. Leakage-Free Cold-Start Variant

The same-review hybrid model obtains the VADER sentiment and TF-IDF features directly from the review for which the Score is being predicted. In a genuine cold-start prediction setting, this information would not be available because the review would not yet exist. Therefore, a second leakage-free hybrid configuration is evaluated using leave-one-out aggregates derived from other reviews associated with the same product.

For each target review, the mean VADER compound score and mean TF-IDF vector are calculated using all other reviews for the same product, with the target review excluded from the aggregation. For products with no other associated reviews, which account for 3,166 of the 100,001 rows (3.2%), the training-set global mean sentiment and content vector are used instead. The aggregated TF-IDF representation is then reduced to 50 components using a TruncatedSVD fitted separately on the aggregated training-row vectors. The SVD collaborative-filtering feature remains unchanged across both configurations. The Random Forest meta-learner is tuned and evaluated using the same canonical split and procedure outlined in Section III-H.

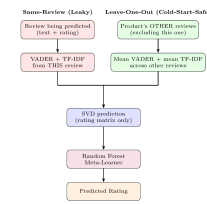


Fig. 3. The two hybrid feature-construction strategies compared in Section IV. The same-review configuration (left) derives VADER sentiment and TF-IDF features from the review whose rating is being predicted. The leave-one-out configuration (right) derives the same features from a leakage-free aggregate of the same product's other reviews, representing information genuinely available before the target rating exists. Both configurations combine their content features with the same SVD prediction before reaching the Random Forest meta-learner.

IV. RESULTS

Table I reports the validation-set grid search used to select each model's hyperparameters (Section III-H).

Model	Configuration	Validation RMSE
SVD	factors=50, epochs=20, lr=0.005, reg=0.02	1.1079
SVD	factors=100, epochs=20, lr=0.005, reg=0.02	1.1095
SVD	factors=100, epochs=30, lr=0.005, reg=0.05	1.0715
SVD	factors=50, epochs=30, lr=0.01, reg=0.05 (selected)	1.0015
Random Forest	100 trees, unlimited depth	0.9570
Random Forest	200 trees, unlimited depth (selected)	0.9565
Random Forest	100 trees, max depth 20	0.9585
Random Forest	200 trees, max depth 20	0.9582

TABLE I
HYPERPARAMETER TUNING (VALIDATION RMSE)

Table II compares the overall test-set error of the tuned SVD-only baseline and the tuned hybrid model on the canonical held-out test split.

Metric	SVD-Only	Hybrid Model
RMSE	1.0068	0.9604
MAE	0.7356	0.5907

TABLE II
OVERALL ERROR COMPARISON (TUNED MODELS)

The hybrid model achieves a 4.61% reduction in RMSE and a 19.71% reduction in MAE relative to the tuned SVD-only baseline. These improvements are lower than those obtained in the earlier untuned comparison, which reported reductions of 11.03% and 27.05%, respectively. Fair tuning substantially strengthens the SVD baseline, lowering its test RMSE from 1.0903 to 1.0068. An earlier version of the evaluation was affected by a train/test leakage issue and reported improvements of 46.75% and 59.16%; these results are therefore not considered valid. The 4.61% RMSE and 19.71% MAE reductions represent the valid comparison between the tuned models using the held-out test set.

Table III presents the error results according to the true star rating.

Rating	N	Hybrid RMSE	Hybrid MAE	SVD RMSE	SVD MAE
1	1165	2.226	1.857	2.437	2.269
2	797	1.604	1.352	1.705	1.585
3	1287	1.006	0.805	0.971	0.856
4	2424	0.609	0.450	0.427	0.336
5	9328	0.639	0.374	0.680	0.559

TABLE III
ERROR BY TRUE RATING CLASS (TUNED MODELS)

The hybrid model achieves lower errors than the SVD baseline for 1, 2, and 5-star reviews, while the SVD model performs better for 3 and 4-star reviews. This pattern remains unchanged following hyperparameter tuning and was also present in the earlier untuned comparison. The consistency of this pattern suggests that the observed differences are associated with the dataset and model behaviour rather than being caused by a particular hyperparameter configuration.

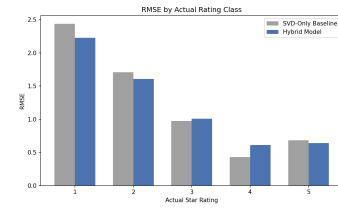


Fig. 4. RMSE by true star rating, SVD-only baseline vs. hybrid model (same-review configuration). The crossover pattern, hybrid ahead at 1, 2, and 5 stars; SVD ahead at 3 and 4 is shown directly.

Table IV summarises the Random Forest feature importance by feature group.

Feature Group	Aggregate Importance
SVD prediction (collaborative)	0.9549
TF-IDF components (content)	0.0423
VADER sentiment	0.0029

TABLE IV
AGGREGATE FEATURE IMPORTANCE (TUNED MODEL)

In the tuned configuration, the SVD prediction’s decision weight within the Random Forest model rose to 95.5%, up from 85.4% in the previous untuned setup. Because the tuned SVD prediction makes a larger impact, the remaining features account for smaller shares: TF-IDF contributes 4.2% and VADER contributes 0.3%. These findings demonstrate that as the core collaborative filtering model gains accuracy, the incremental value provided by review-text features diminishes.

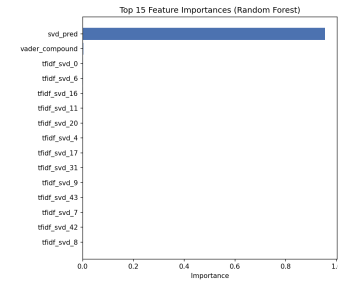


Fig. 5. Top 15 individual feature importances (Random Forest, same-review configuration). Beyond `svd_pred`, every remaining feature contributes a small fraction of total importance, illustrating how dominant the collaborative signal is once tuned. The SVD prediction dominates the feature contributions, while the remaining features account for relatively small proportions of the overall importance.

Table V reports Precision@K for the two evaluation configurations described in Section III-I.

Metric	Coverage	SVD-Only	Hybrid
Precision@3 (all users)	80.364% (9196/11443)	0.9749	0.9754
Precision@10 (≥ 10 items)	0.07% (8/11443)	0.9000	0.9125

TABLE V
PRECISION@K COMPARISON (TUNED MODELS)

Evaluating Precision@3 across 80.4% of test users reveals virtually identical performance between the two models—0.9749 for the SVD-only setup and 0.9754 for the hybrid approach. This near-ceiling outcome stems heavily from the dataset’s 62% positive-rating skew, rather than ranking capacity alone. Conversely, Precision@10 shows a slight edge for the hybrid model over SVD (0.9125 vs. 0.9000); however, because this metric reflects just 8 users, it offers limited standalone evidence.

Leakage-Free Cold-Start Comparison: Table VI compares the tuned SVD baseline with the same-review hybrid and the leakage-free leave-one-out hybrid.

Metric	SVD-Only	Same-Review Hybrid	Leave-One-Out Hybrid
RMSE	1.0068	0.9604 (-4.61%)	0.9998 (-0.69%)
MAE	0.7356	0.5907 (-19.71%)	0.6224 (-15.40%)
SVD feature importance	-	95.49%	95.62%
Content feature importance (TF-IDF + sentiment)	-	4.23% + 0.29%	3.85% + 0.54%

TABLE VI

SVD-ONLY VS. SAME-REVIEW HYBRID VS. LEAKAGE-FREE HYBRID

Excluding same-review information sharply reduces the RMSE gain from 4.61% down to 0.69%, demonstrating that the performance boost in the same-review setup relies heavily on direct access to the target review text. In contrast, the MAE improvement drops more moderately—from 19.71% to 15.40%, which indicates that aggregated sentiment and content from a product’s other reviews still offer predictive power past the collaborative filtering baseline.

This divergence between metrics stems from how they calculate errors: RMSE penalizes larger prediction errors more heavily, whereas MAE weights all individual errors equally. Consequently, the specific sentiment of the target review helps correct occasional large errors more effectively, while aggregating review text at the product level smooths out individual sentiment fluctuations across multiple reviews.

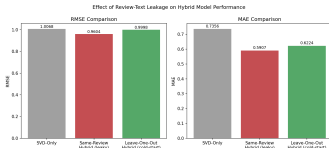


Fig. 6. RMSE and MAE across all three configurations: SVD-only baseline, same-review (leaky) hybrid, and leave-one-out (cold-start-safe) hybrid.

Table VII confirms the rating-class pattern under the leakage-free evaluation.

Rating	N	Cold-Start Hybrid RMSE	Cold-Start Hybrid MAE	SVD RMSE	SVD MAE
1	1165	2.327	1.997	2.437	2.269
2	797	1.683	1.442	1.705	1.585
3	1287	1.041	0.837	0.971	0.856
4	2424	0.606	0.433	0.427	0.336
5	9328	0.667	0.400	0.680	0.559

TABLE VII

ERROR BY TRUE RATING CLASS (LEAKAGE-FREE HYBRID)

The leakage-free hybrid model maintains its advantage over SVD for 1, 2, and 5-star reviews, while remaining less accurate

on 3 and 4-star reviews. Consequently, this same rating-class pattern persists across both feature-construction approaches.

Precision@K shows virtually no change in the leakage-free setup. Precision@3 holds at 0.9749 across both configurations, and Precision@10 remains identical at 0.9000—a figure based on a sample of just 8 users. This minimal shift aligns with how heavily the dataset’s rating imbalance impacts ranking-based evaluation metrics.

V. DISCUSSION AND LIMITATIONS

Three key methodological considerations shape the interpretation of these findings.

First, the same-review hybrid model derives its VADER sentiment and TF-IDF content features from the exact review whose rating score is being predicted. However, true cold-start recommendation systems must estimate ratings before a user writes the review, ruling out the use of that target text as an input. Section III-J and Table VI address this constraint via a leakage-free, leave-one-out setup. Substituting same-review features with aggregated data from the product’s other reviews cuts the RMSE improvement from 4.61% to 0.69%. The MAE gain drops less dramatically, from 19.71% to 15.40%, showing that product-level review aggregation still adds predictive value beyond SVD under leakage-free conditions.

Second, severe per-user data sparsity impacts how ranking-based metrics should be understood. Because the median test user has under two test items, the dataset falls short of typical top-K evaluation thresholds. As a result, Precision@3 shows a near-ceiling effect driven by the dataset’s skewed ratings, whereas Precision@10 applies to only a tiny subset of users. Future research using this dataset might instead adopt Recall@K, NDCG with lower K values, or restrict evaluation to highly active users.

Third, applying rigorous hyperparameter tuning significantly shifts the dynamic between the hybrid model and the SVD baseline. Initial untuned testing showed an 11.03% RMSE advantage for the hybrid model; however, tuning both models under identical validation setups shrunk that gain to 4.61%. This drop occurs because the tuned SVD baseline achieves much higher accuracy. This underscores the necessity of comparing hybrid models against well-tuned baselines, as untuned baselines artificially inflate the perceived utility of auxiliary features.

Feature-importance results reinforce this point. Upon tuning, the Random Forest model increases its decision weight for the SVD prediction from 85.4% to 95.5%, leaving the text-based features to supply only a minor incremental signal on top of the strengthened collaborative filtering foundation.

In conclusion, while these results remain insightful, the hybrid model’s value is best viewed as targeted rather than expansive or universal. Across both feature-building methods, leakage-free performance gains cluster primarily in the 1, 2, and 5-star rating categories, while ranking advantages measured by Precision@K remain limited by user sparsity and rating bias. Ultimately, the evaluation offers a precise picture

of where and to what degree, review-based features augment collaborative filtering under these specific test parameters.

VI. ETHICAL AND SOCIAL CONSIDERATIONS

This study utilizes the publicly available, anonymous Amazon Fine Food Reviews dataset, featuring pseudonymous user IDs that prevent direct identification. The research required no additional personal data collection, human subject recruitment, or gathering of new individual information. Therefore, the work adheres to the data minimization and privacy guidelines outlined in both the ACM Code of Ethics and the BCS Code of Conduct.

The study's findings highlight two specific fairness considerations:

- **Popularity Bias:** A sharp positive-rating skew (where 62% of reviews are five stars), paired with long-tailed distributions for user and product activity (detailed in Section III-B), risks favoring high-activity users and popular items. Section IV notes that the SVD baseline struggles most under sparse user and item conditions. This aligns with standard recommender system popularity bias, where models trained on skewed interactions can perpetuate existing trends instead of offering equal exposure to less active products.
- **Cold-Start Assistance:** The leakage-free, leave-one-out setup from Section III-J presents a useful tactic for items with sparse review histories by deriving content features from an item's existing reviews rather than depending on extensive user-rating profiles. As shown in Section IV, this strategy yields modest yet measurable improvements under cold-start scenarios, allowing product-level data to supplement collaborative filtering when low interaction volume hinders SVD performance.

Additionally, feature reliability introduces risks. Because VADER scores stem straight from raw review text, any underlying noise, bias, or deceptive text directly skews the derived sentiment and content inputs. On e-commerce sites, non-genuine, paid, or coordinated reviews pose a significant source of noise. Since this study omits fake-review detection mechanisms, real-world implementations must account for this vulnerability, as manipulated text would directly taint the features detailed in Sections III-D and III-E.

To conclude, the practical utility of this approach remains compelling. Rating-prediction and recommendation frameworks lower consumer search friction when evaluating vast product catalogs. Moreover, the feature-importance analysis in Table IV offers clear insight into the balance between collaborative and content signals providing superior interpretability compared to the end-to-end neural models referenced in Section II-C. For platform managers and smaller merchants, integrating review text adds valuable product-level context without demanding the massive interaction logs required by unassisted collaborative filtering.

VII. CONCLUSION AND FUTURE WORK

This paper presented a hybrid recommender system combining SVD-based collaborative filtering, VADER sentiment scoring, and dimensionality-reduced TF-IDF content features through a Random Forest meta-learner. Evaluation relied on a stratified sample of 100,001 Amazon Fine Food Reviews with a single canonical 70/15/15 train/validation/test split shared across all models. Hyperparameters for both models were selected through grid search using the validation split. Under the same-review feature configuration, the hybrid model reduced RMSE by 4.61% and MAE by 19.71% compared with the fairly tuned SVD-only baseline. Because the same-review configuration derives content features from the review associated with the predicted rating, a second leakage-free variant was evaluated using leave-one-out product-level aggregates from other reviews. Under this cold-start-safe evaluation, the RMSE improvement decreased substantially to 0.69%, while the MAE improvement remained at 15.40%. These results show that genuinely available review content continues to provide predictive value beyond collaborative filtering, although the improvement is modest.

This direct comparison represents a central methodological contribution of the study because it demonstrates how much of the apparent advantage of a hybrid model depends on information that would not be available at the actual time of recommendation. The comparison also identifies the portion of the improvement that remains after same-review information is removed through a fair, leakage-free evaluation.

Three additional factors are important when interpreting the findings:

- **Sparsity Constraints:** Strong per-user sparsity in the dataset limits the interpretation of Precision@K at conventional K values.
- **Tuning Impacts:** Fair tuning of the SVD baseline substantially reduces the apparent advantage of the hybrid model compared with an untuned comparison, highlighting the importance of tuning both models when evaluating a baseline against a proposed approach.
- **Rating Patterns:** Per-rating-class results show a consistent pattern across both feature-construction strategies, with the hybrid approach performing better for extreme ratings and the SVD baseline performing better for mid-range ratings. The consistency of this pattern under both configurations suggests that the behavior is associated with the data rather than being an artifact of a single model configuration.

Feature-importance analysis further supports this interpretation. The Random Forest assigns 95.5% of its decision weight to the SVD prediction after tuning, compared with 85.4% before tuning. This indicates that the text-derived features provide a modest additional signal on top of a well-tuned collaborative filtering model rather than representing the dominant source of predictive information.

Future work should consider Recall@K or NDCG at smaller K values that better reflect the dataset's sparsity. A broader hy-

perparameter search could also be explored beyond the small grids used in the current study. Further investigation could replace the Random Forest meta-learner with neural fusion architectures [12], [13], increase TF-IDF dimensionality, or use transformer-based text embeddings instead of TF-IDF. Such approaches could also be combined with the leave-one-out aggregation strategy introduced in this study.

REFERENCES

- [1] J. Bobadilla, F. Ortega, A. Hernando, and A. Gutiérrez, “Recommender systems survey,” *Knowledge-Based Systems*, vol. 46, pp. 109–132, 2013.
- [2] F. Ricci, L. Rokach, and B. Shapira, *Recommender Systems Handbook*, 3rd ed. Springer, 2022.
- [3] Y. Koren, R. Bell, and C. Volinsky, “Matrix factorization techniques for recommender systems,” *Computer*, vol. 42, no. 8, pp. 30–37, 2009.
- [4] A. Paterek, “Improving regularized singular value decomposition for collaborative filtering,” in *Proc. KDD Cup and Workshop*, 2007, pp. 5–8.
- [5] R. Salakhutdinov and A. Mnih, “Probabilistic matrix factorization,” in *Advances in Neural Information Processing Systems (NIPS)*, vol. 20, 2007, pp. 1257–1264.
- [6] J. L. Herlocker, J. A. Konstan, L. G. Terveen, and J. T. Riedl, “Evaluating collaborative filtering recommender systems,” *ACM Trans. Inf. Syst.*, vol. 22, no. 1, pp. 5–53, 2004.
- [7] J. A. Konstan and J. Riedl, “Recommender systems: from algorithms to user experience,” *User Modeling and User-Adapted Interaction*, vol. 22, no. 1–2, pp. 101–123, 2012.
- [8] J. McAuley and J. Leskovec, “Hidden factors and hidden topics: Understanding rating dimensions with review text,” in *Proc. 7th ACM Conference on Recommender Systems (RecSys)*, 2013, pp. 165–172.
- [9] J. McAuley and J. Leskovec, “From amateurs to connoisseurs: Modeling the evolution of user expertise through online reviews,” in *Proc. 22nd International Conference on World Wide Web (WWW)*, 2013.
- [10] X. He, T. Chen, M.-Y. Kan, and X. Chen, “TriRank: Review-aware explainable recommendation by modeling aspects,” in *Proc. 24th ACM International Conference on Information and Knowledge Management (CIKM)*, 2015, pp. 1661–1670.
- [11] Y. Tan, M. Zhang, Y. Liu, and S. Ma, “Rating-boosted latent topics: Understanding users and items with ratings and reviews,” in *Proc. 25th International Joint Conference on Artificial Intelligence (IJCAI)*, 2016, pp. 2640–2646.
- [12] X. He, L. Liao, H. Zhang, L. Nie, X. Hu, and T.-S. Chua, “Neural collaborative filtering,” in *Proc. 26th International Conference on World Wide Web (WWW)*, 2017, pp. 173–182.
- [13] R. Catherine and W. Cohen, “TransNets: Learning to transform for recommendation,” in *Proc. 11th ACM Conference on Recommender Systems (RecSys)*, 2017, pp. 288–296.
- [14] S. Zhang, L. Yao, A. Sun, and Y. Tay, “Deep learning based recommender system: A survey and new perspectives,” *ACM Computing Surveys*, vol. 52, no. 1, pp. 1–38, 2019.
- [15] Y. Deldjoo, Z. He, J. McAuley, A. Korikov, S. Sanner, A. Ramisa, R. Vidal, M. Sathiamoorthy, A. Kasirzadeh, and S. Milano, “A review of modern recommender systems using generative models (Gen-RecSys),” *arXiv preprint arXiv:2404.00579*, 2024.
- [16] C. J. Hutto and E. Gilbert, “VADER: A parsimonious rule-based model for sentiment analysis of social media text,” *Proceedings of the International AAAI Conference on Web and Social Media (ICWSM)*, vol. 8, no. 1, pp. 216–225, 2014.
- [17] T. Silveira *et al.*, “How good is your recommender system? A survey on evaluation,” *International Journal of Machine Learning and Cybernetics*, 2019.
- [18] B. M. Sarwar, G. Karypis, J. A. Konstan, and J. Riedl, “Application of dimensionality reduction in recommender system – A case study,” in *ACM WebKDD Workshop*, 2000.
- [19] D. Jurafsky and J. H. Martin, *Speech and Language Processing*, 3rd ed. Stanford University, 2023, draft.
- [20] Y. Goldberg, *Neural Network Methods for Natural Language Processing*. Morgan & Claypool, 2017.
- [21] P. M. LeBlanc, D. Banks, L. Fu, M. Li, Z. Tang, and Q. Wu, “Recommender systems: A review,” *Journal of the American Statistical Association*, vol. 119, no. 545, pp. 773–785, 2024.
- [22] E. Hasan, M. Rahman, C. Ding, J. X. Huang, and S. Raza, “Review-based recommender systems: A survey of approaches, challenges and future perspectives,” *ACM Computing Surveys*, vol. 58, no. 1, 2025.
- [23] I. Portugal, P. Alencar, and D. Cowan, “The use of machine learning algorithms in recommender systems: A systematic review,” *arXiv preprint arXiv:1511.05263*, 2015.